

AREAL , DeepSeek Papers: RL Case Study + Optimizations

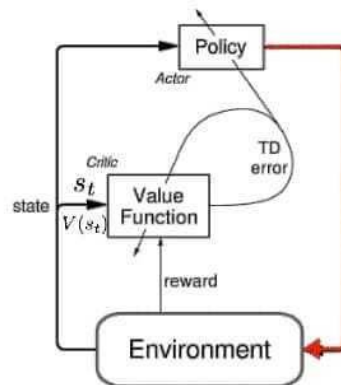
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RL Background



Actor-Critic



- Actor: decides which action to take
- Critic: tells the actor how good its action was and how it should adjust

(Figure from Sutton & Barto, 1998)

RL Background

Critic loss function

$$L_c(\theta_c) = \left(\overbrace{(r_t + \gamma V_{\theta_c}(s_{t+1})) - V_{\theta_c}(s_t)}^{\text{TD error / Advantage}} \right)^2$$

Parameters of the critic

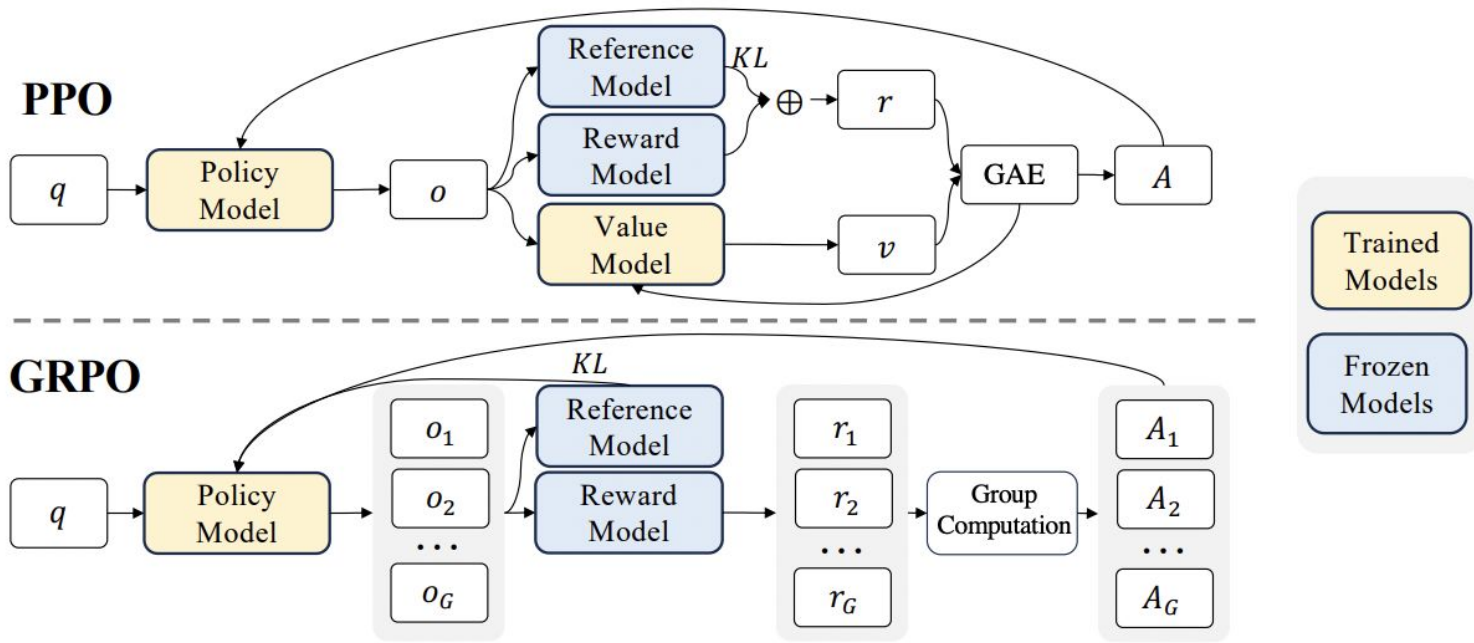
PPO

- Problems with Actor-Critic
 - Policy collapses if critic invalid, training explodes, probs change lots
 - Small steps lead to slow learning
 - Big steps lead to unstable updates
- Instead of evaluating an action, compare likelihood of action between new and old policy
- Ratio of how much to update: $\pi_{\text{new}}(a_t | s_t) / \pi_{\text{old}}(a_t | s_t)$
- Clips the ratio

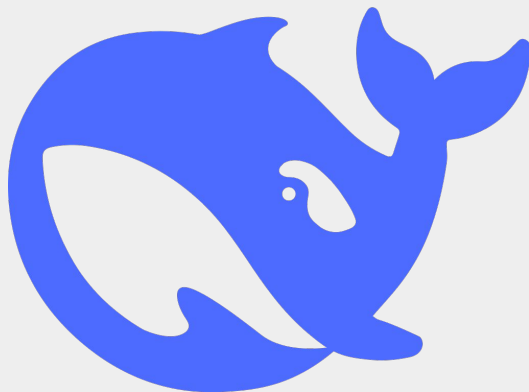
GRPO

- Eliminate the need for the critic (expensive and complicated)
- Sample a bunch of outputs
- Estimate advantage based on how good the output is compared to the rest
- Update based on this advantage

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}.$$



DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning



Reasoning Problems

- Prior Work: **Chain of Thought (CoT) supervised learning**
 - Requires many human annotations
 - Step by step reasoning processes
 - Limited by human reasoning
 - Very expensive
 - Supervised Fine Tuning (SFT) imitates

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

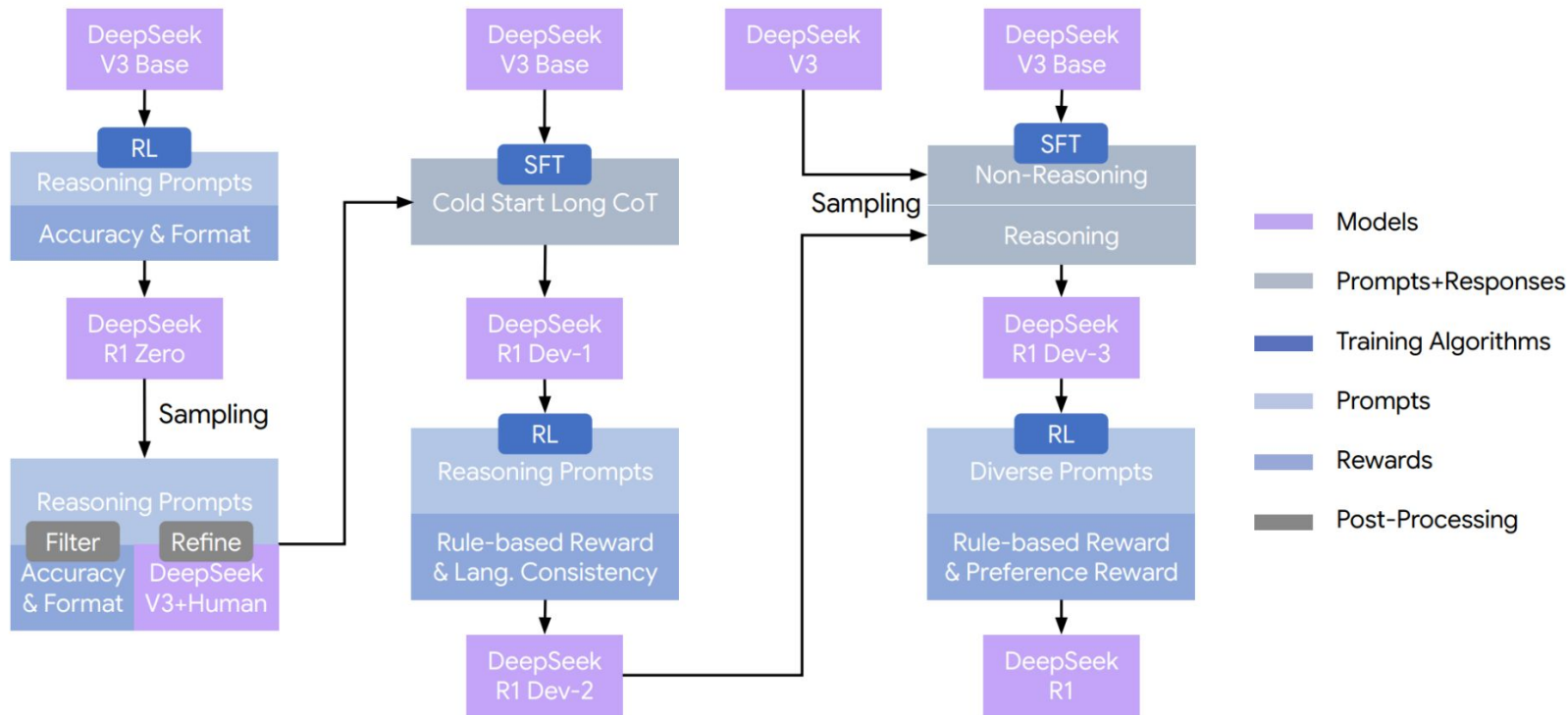
DeepSeek R1 Solution

- **Reinforcement Learning (RL)**
 - Optimizes for correctness (accuracy of answer + format)
 - No need for human labels
 - Avoid reward hacking
 - Reasoning follows...
- SFT to format and follow instruction
- Distillation to smaller models
- Result: LLM with reasoning capability
- Impressed many without needing as much!

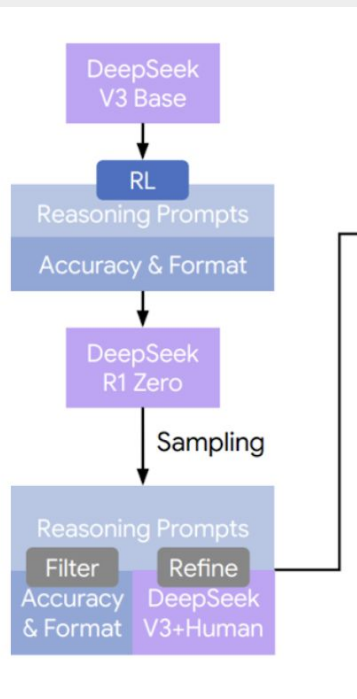
Reasoning Not Enough

- Initial reward is only **accuracy + format**
- Just RL leads to reasoning not human comprehensible
- Lots of language mixing because base model uses Chinese and English
- SFT to format and follow instructions
- Need to translate reasoning into human readable language

1. 今天我在 library 里学习 machine learning, 感觉 concepts 终于开始 make sense 了。
2. 我们晚上一起吃 dinner 吧, maybe order some dumplings 然后聊聊天。

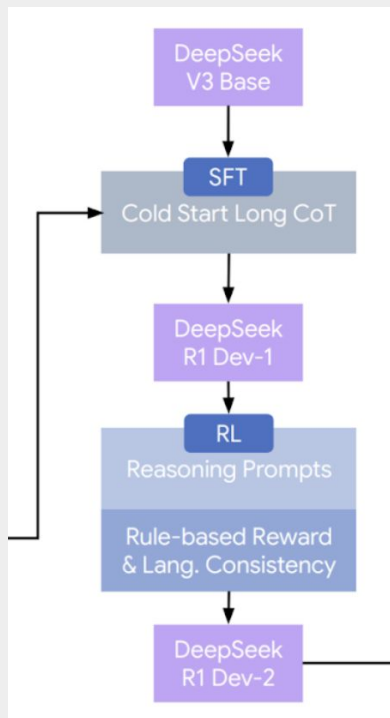


RL Stage 0: Reasoning Emergence



- Start: DeepSeek V3 Base
- RL(GRPO) on reasoning prompts only
- Reward Signals: Correctness + Format
- No SFT/Reward model or human CoT examples
- Results: Emergent (+ messy) CoT with self-verification

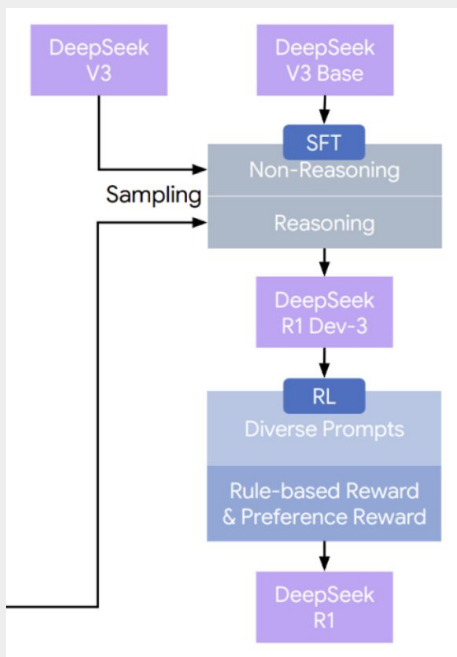
RL Stage 1: Reasoning Stabilization



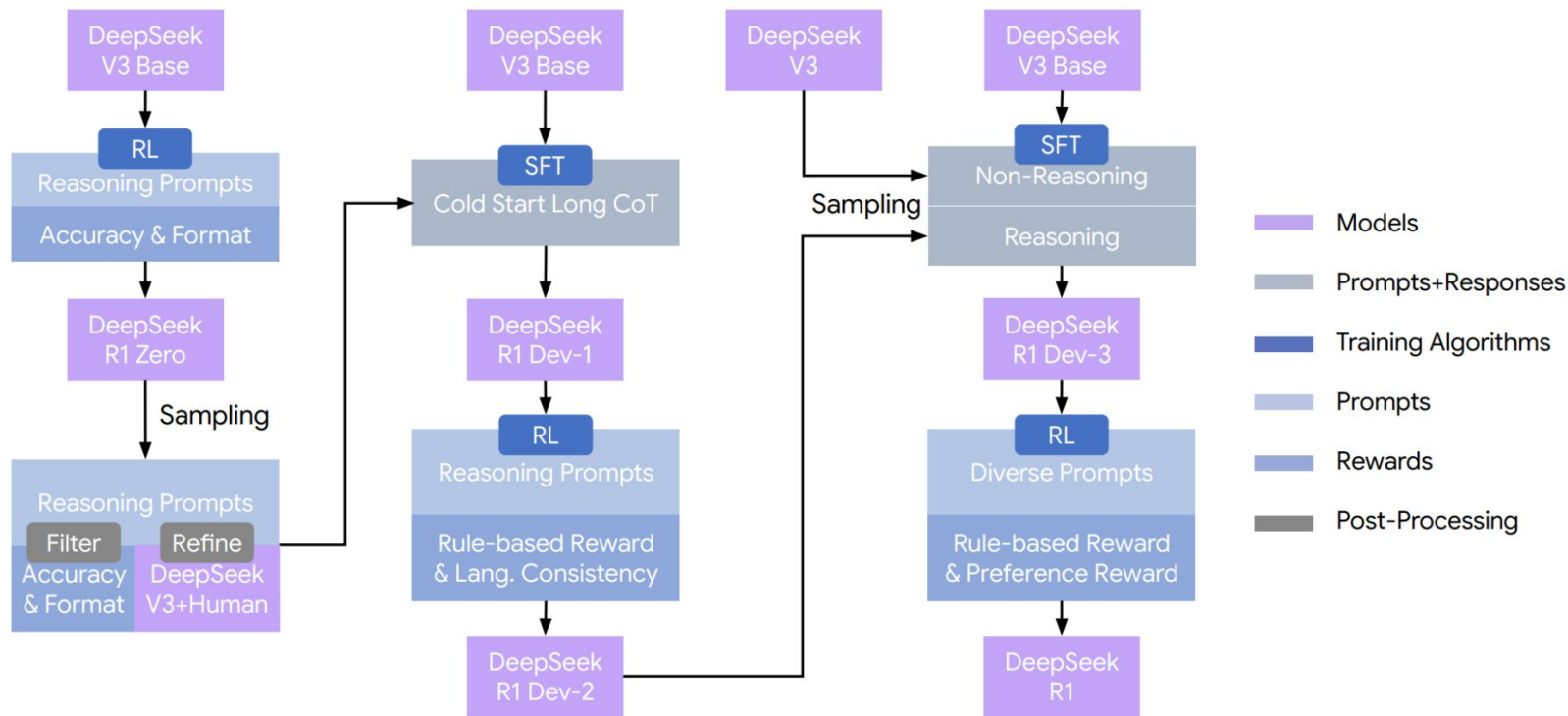
- Small amount of CoT Data added(clean + readable)
- SFT improves readability not reasoning strategy
- RL(GRPO) continues on reasoning prompts
- Add Reward to ensure language consistency
- Small Drop in reasoning performance -> Readability + Stability



RL Stage 2: User Alignment



- SFT on both reasoning/non reasoning data
- Distills strong reasoning into training examples
- Final RL stage with diverse prompts
- Rewards: Rule based + helpfulness and safety
- End Result: Reasoning + follows instructions



Benchmark (Metric)		R1-Zero	R1-Dev1	R1-Dev2	R1-Dev3	R1
English	MMLU (EM)	88.8	89.1	91.2	91.0	90.8
	MMLU-Redux (EM)	85.6	90.0	93.0	93.1	92.9
	MMLU-Pro (EM)	68.9	74.1	83.8	83.1	84.0
	DROP (3-shot F1)	89.1	89.8	91.1	88.7	92.2
	IF-Eval (Prompt Strict)	46.6	71.7	72.0	78.1	83.3
	GPQA Diamond (Pass@1)	75.8	66.1	70.7	71.2	71.5
	SimpleQA (Correct)	30.3	17.8	28.2	24.9	30.1
	FRAMES (Acc.)	82.3	78.5	81.8	81.9	82.5
	AlpacaEval2.0 (LC-winrate)	24.7	50.1	55.8	62.1	87.6
	ArenaHard (GPT-4-1106)	53.6	77.0	73.2	75.6	92.3
Code	LiveCodeBench (Pass@1-COT)	50.0	57.5	63.5	64.6	65.9
	Codeforces (Percentile)	80.4	84.5	90.5	92.1	96.3
	Codeforces (Rating)	1444	1534	1687	1746	2029
	SWE Verified (Resolved)	43.2	39.6	44.6	45.6	49.2
	Aider-Polyglot (Acc.)	12.2	6.7	25.6	44.8	53.3
Math	AIME 2024 (Pass@1)	77.9	59.0	74.0	78.1	79.8
	MATH-500 (Pass@1)	95.9	94.2	95.9	95.4	97.3
	CNMO 2024 (Pass@1)	88.1	58.0	73.9	77.3	78.8
Chinese	CLUEWSC (EM)	93.1	92.8	92.6	91.6	92.8
	C-Eval (EM)	92.8	85.7	91.9	86.4	91.8
	C-SimpleQA (Correct)	66.4	58.8	64.2	66.9	63.7

Reasoning in Practice

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...



Summary

- Reinforcement Learning: efficient and powerful way for LLM reasoning
- Balance SFT + RL stages to output human readable reasoning
 - SFT for readability and instruction-following
 - RL for reasoning and then later human preferences
- Relies on a reliable verifier and sufficient computation for RL

RL is becoming more important for LLMs

Limitations

- Structured Output and Tool Use
- Token Efficiency (excessive reasoning, overthinking)
- Few-shot Prompting Degrades Performance
- Limited Improvement in SWE Tasks (Long Eval times)
- Defining Rewards for Problems
- **Training Efficiency of Synchronous RL**

Training Efficiency of Synchronous RL

- Micro batches have hugely different generation times
- Synchronous: Need outputs from a given iteration before update step
- Imbalanced batch costs -> Significant idle GPU time
 - (Similar problem as WLB-LLM)
- Reasoning tokens scale with problem complexity, not input size

AREAL: A Large-Scale Asynchronous Reinforcement Learning System for Language Reasoning

Inefficiency From Non-Uniform Decoding Length:

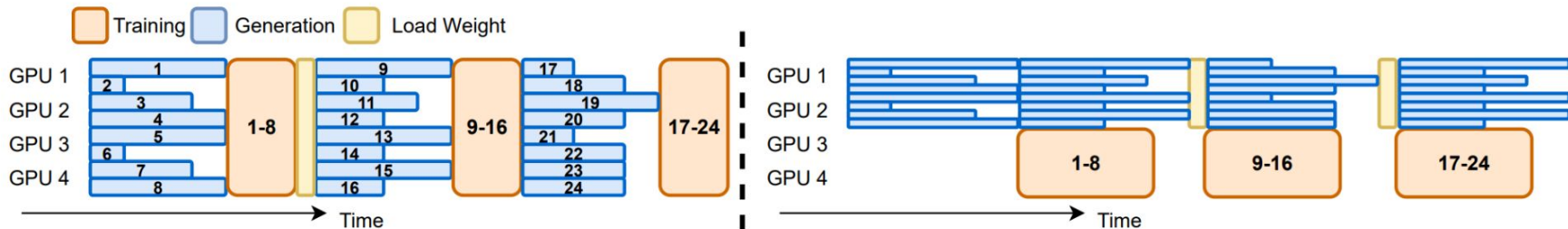


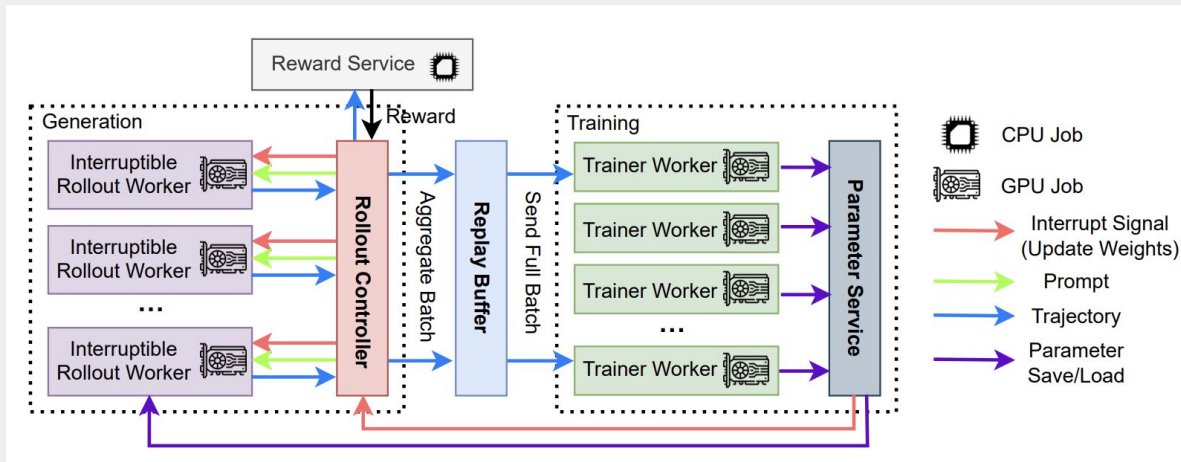
Figure 1: Execution timeline of a synchronous (left) and a one-step overlap (right) RL system showing underutilized inference devices.

Poor Scalability in Synchronous RL

- More GPUs -> Smaller Batches -> Worse Performance
- Small batches fail to saturate SMs
- I/O bound rather than compute bound

AReaL: Fully Asynchronous RL Training Framework

- Interruptible Rollout Worker
 - Generates Response
 - Updates Weights (interrupts generation tasks)
- Reward Service
 - Evaluates accuracy
- Trainer Workers
 - Perform PPO Updates
- Rollout Controller
 - Orchestration



Interruptible Generation

- Nodes immediately start the next batch's generation task
- Weights are updated only on interrupts
- Generation is resumed with the updated weights
- KV-Cache recompute cost when resuming mid-batch

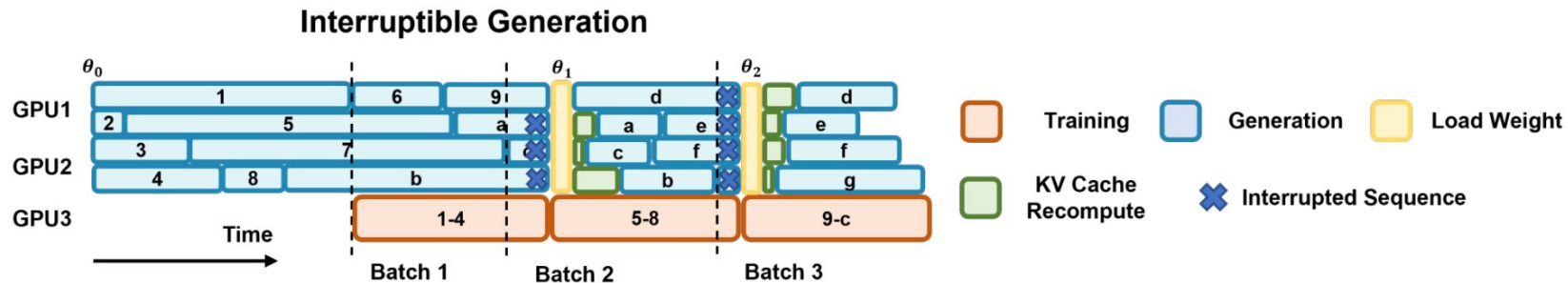


Figure 3: Illustration of generation management in AREAL. Vertical lines show the ready time for the next step training. Blue crosses show the interrupted requests when new parameters arrive.

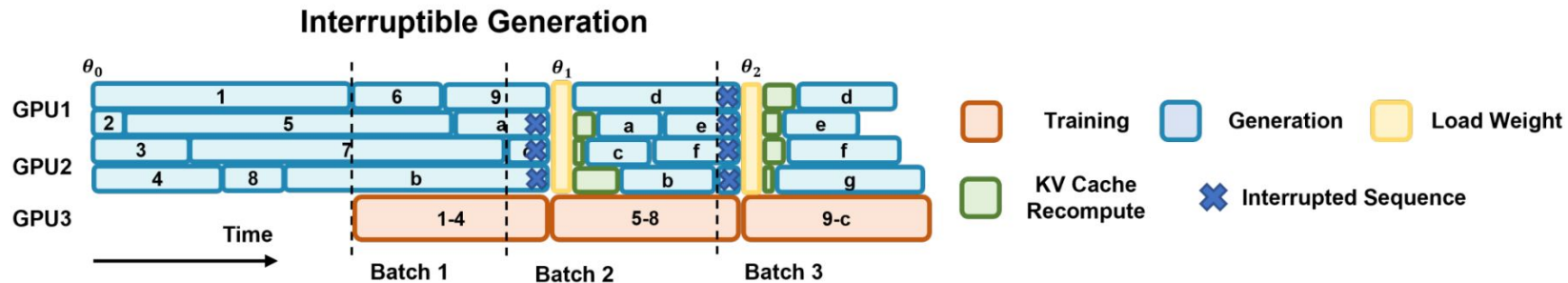
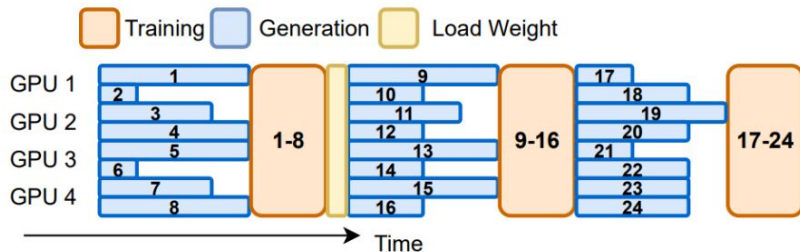


Figure 3: Illustration of generation management in AREAL. Vertical lines show the ready time for the next step training. Blue crosses show the interrupted requests when new parameters arrive.



Challenges

- Stale training data
 - Inconsistency leads to degraded performance
- Batch may contain data from multiple policy version, violating PPO expectations
- Even individual segments may have tokens from multiple policies

Staleness-Aware Training

- Define a *maximum permitted staleness* (η) for each training batch
- Implemented as a rate limiter by dynamically controlling throughput to generation server

$$\lfloor (N_r - 1)/B \rfloor \leq i + \eta.$$

Decoupled PPO Objective

- Standard PPO assumes one “old” policy
 - AReal breaks this due to asynchronous nature

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{q \sim \mathcal{D}, a_t \sim \pi_{\text{old}}(\cdot | q, a_{<t})} \left[\sum_{t=1}^H \min \left(u_t(\theta) \hat{A}(s_t, a_t), \text{clip} \left(u_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}(s_t, a_t) \right) \right], \quad (2)$$

$$u_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{old}}(a_t | s_t)}$$

- Must redefine policy notions in order to prevent bias towards older policies

Decoupled PPO Objective

- Disentangle policy into *behavior* and *proximal* policies

π_{behav}	Policy (or policies*) used to generate trajectory
π_{prox}	Recent “trusted” policy used as a regularization anchor
π_{θ}	Global policy

*Inconsistent policy versions across a trajectory proven to maintain equivalence to a single behavioral policy

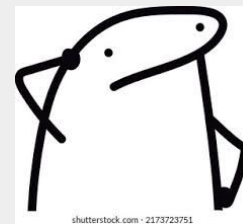
Decoupled PPO Objective

- Use proximal policy for regularization to prevent bias towards stale versions during policy updates

$$J(\theta) = \mathbb{E}_{q \sim \mathcal{D}, a_t \sim \pi_{\text{behav}}} \left[\sum_{t=1}^H \min \left(\underbrace{\frac{\pi_{\theta}}{\pi_{\text{behav}}}}_{\text{Importance Ratio}} \hat{A}_t, \overbrace{\frac{\pi_{\text{prox}}}{\pi_{\text{behav}}} \text{clip} \left(\underbrace{\frac{\pi_{\theta}}{\pi_{\text{prox}}}}_{\text{Trust Region Center}}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t}_{\text{Importance Ratio}} \right) \right] \quad (4)$$

$$= \mathbb{E}_{q \sim \mathcal{D}, a_t \sim \pi_{\text{behav}}} \left[\sum_{t=1}^H \frac{\pi_{\text{prox}}}{\pi_{\text{behav}}} \min \left(u_t^{\text{prox}}(\theta) \hat{A}_t, \text{clip} \left(u_t^{\text{prox}}(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (5)$$

$$u_t^{\text{prox}}(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{prox}}(a_t | s_t)}$$



Experiment Results

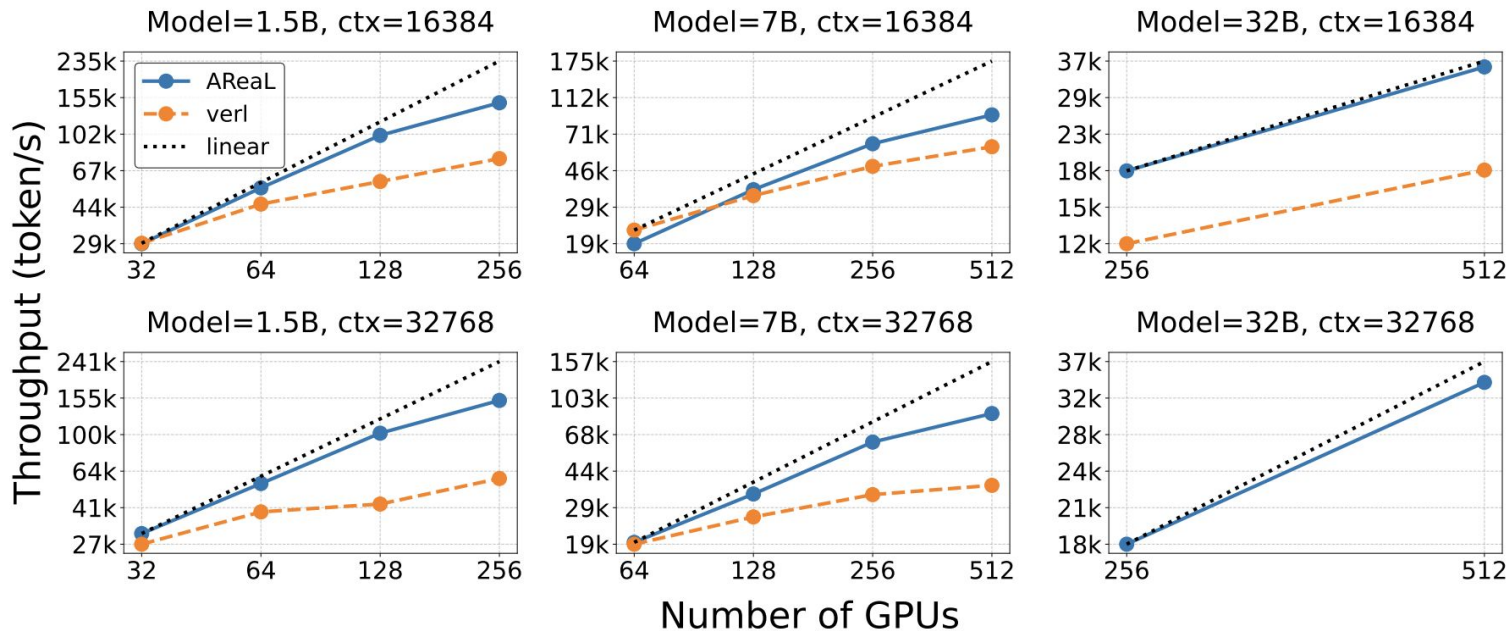
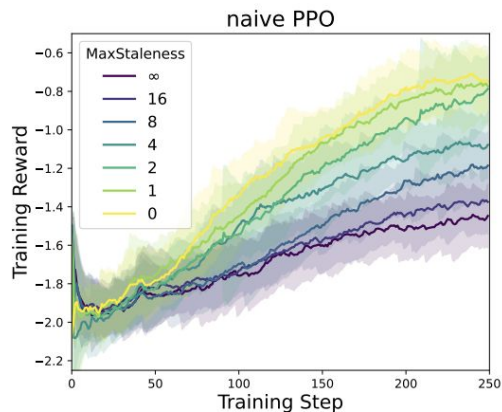
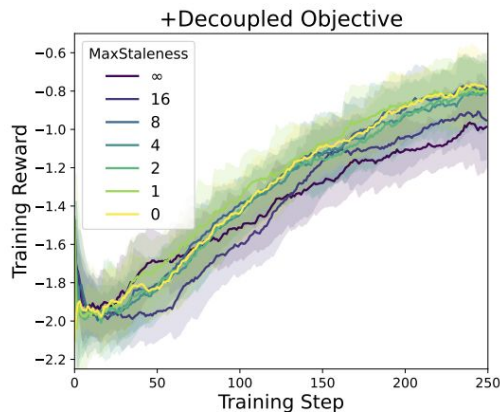


Figure 4: The strong scaling trend. Dotted lines indicate ideal linear scaling. verl consistently encounters OOM with 32k context length and the 32B model so the data points are missing.

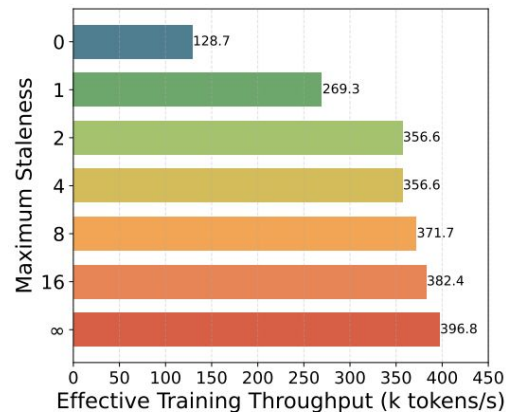
Model	AIME24 ↑	# Nodes	PPO Steps	Training Hours ↓
1.5B basemodel	29.3	-	-	-
w/ VeRL	43.1*	16	250	33.6
w/ Sync.AReaL	42.0	16	250	41.0
w/ AReaL (ours)	42.2	16	250	14.8
7B basemodel	54.3	-	-	-
w/ VeRL	-	24	250	52.1
w/ Sync.AReaL	63.0	24	250	57.7
w/ AReaL (ours)	63.1	24	250	25.4
Model	LiveCodeBench ↑	# Nodes	PPO Steps	Training Hours ↓
14B basemodel	53.4	-	-	-
w/ VeRL	57.9*	32	80	44.4
w/ Sync.AReaL	56.7	32	80	48.8
w/ AReaL (ours)	58.1	32	80	21.9
32B basemodel	57.4	-	-	-
w/ VeRL	-	48	60	46.4
w/ Sync.AReaL	61.2	48	60	51.1
w/ AReaL (ours)	61.0	48	60	31.1



(a) Learning curves with naive PPO.



(b) Learning curves with eq. (5).

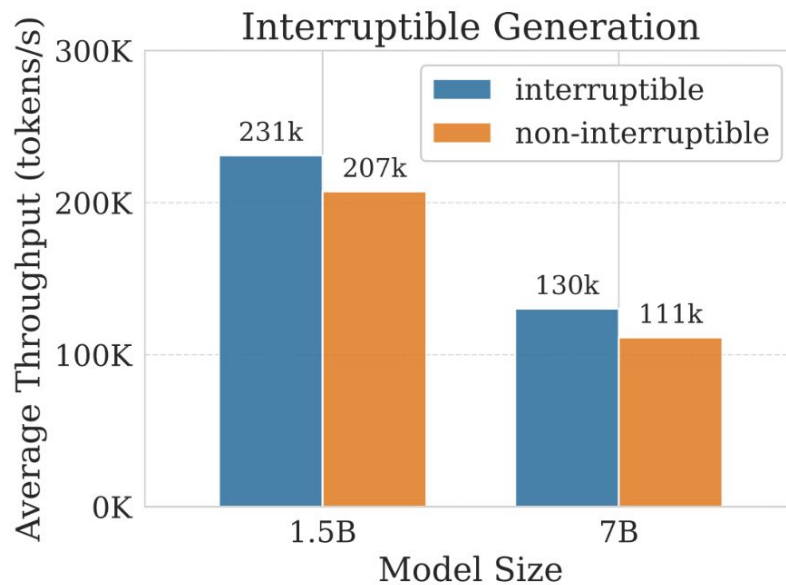


(c) Effective training throughput.

Figure 5: Ablation studies of the decoupled PPO objective and staleness control with a 1.5B model on math reasoning tasks. Both algorithmic choices are essential. With a moderate staleness value and the decoupled objective, training progress can be accelerated by over $2\times$ while maintaining final evaluation performance.

Table 2: Evaluation scores when varying data staleness, comparing performance with and without the decoupled objective. Numbers within ± 1 of the oracle score are underlined.

Max.Stale.	AIME24		AIME25		AMC23		MATH 500	
	W/o	With	W/o	With	W/o	With	W/o	With
0 (Oracle)	42.0		32.9		84.4		89.2	
1	<u>41.8</u>	<u>42.1</u>	30.7	<u>31.9</u>	<u>83.3</u>	<u>85.2</u>	<u>89.9</u>	<u>89.8</u>
2	40.0	<u>41.8</u>	<u>32.1</u>	<u>32.5</u>	82.3	<u>84.3</u>	<u>89.6</u>	<u>89.6</u>
4	23.3	<u>42.2</u>	23.1	<u>32.0</u>	58.5	<u>85.1</u>	66.9	<u>89.5</u>
8	35.7	<u>41.0</u>	27.8	<u>31.1</u>	81.2	82.9	87.8	<u>89.2</u>
16	35.8	38.7	26.2	<u>32.5</u>	78.4	83.2	87.4	<u>89.1</u>
∞	34.0	36.9	26.9	29.9	79.4	81.0	87.1	88.1



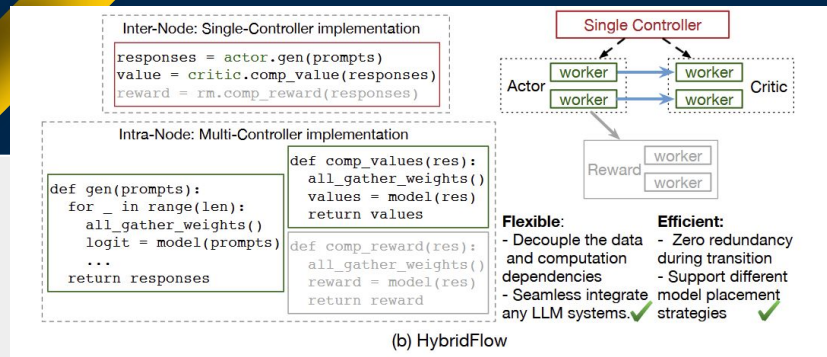
(b) Ablation study of interruptible generation.

Limitations

- **Bounded staleness as a hyperparameter**
 - Impact of staleness can be system dependent and tuning the hyperparameter may be non-trivial
- **Increased system complexity**
 - High traffic between new system components
 - Could result in scalability issues
- **Not necessarily extendable to other RL paradigms**
 - Only focuses on PPO-based optimization

Optional Reading: HybridFlow

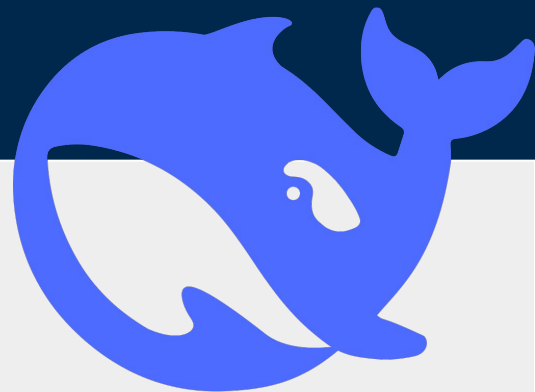
HybridFlow



- Identifies RLHF as a complex, multi-model workflow
- Proposes a **hybrid** programming model, splitting workflow control into two controllers
 - Centralize control *between* models, decentralize execution *within* models
 - Single Controller (Workflow level)
 - Orchestrates the RLHF dataflow, data movement & resharding, etc.
 - Multi-Controller Execution (Model level)
 - Utilizes inter-model parallelism techniques (DP, TP, CP, PP)

Future Research

- Combine HybridFlow and AReaL together
 - HybridFlow controls more macro-level orchestration and AReaL controls more fine grained generation and training scheduling
 - Obvious issues with the conflicting synchronous semantics upheld by the HybridFlow workflow and asynchronous nature of AReaL
- Asynchronous training for other RL paradigms (e.g. GRPO)
- Improve token efficiency by incorporating reasoning length into the reward function



Conclusion

- Deepseek found that RL is a powerful post-training tool to promote model reasoning, tackle challenging problem sets
- Synchronous RL poses throughput concerns
- AReaL introduces an asynchronous mechanism to support throughput while maintaining model quality

Thank You!

Questions?